

Organisational definitions

A ledger of definitions Pearu has approved as **complete**, for the terms `definitions.md` marks *organisational* — those whose content involves persons, obligations, or history. Its purpose is to finish the translation of the whitepaper: `definitions.md` holds what the machinery does; this ledger holds what organisations add around it; the two together should leave no whitepaper term untranslated.

Governance. The rules, clarifications, and entry format of `../.. / RULES.md` govern this file unchanged, as they do `definitions.md`. Nothing enters without explicit per-entry approval.

Prefixes. Primitives are `Q1`, `Q2`, ... and definitions are `O1`, `O2`, ..., so that citations never collide with `definitions.md`'s `Pn/Dn`. The citation convention carries over: `Qn[term]`, `On[term]` — for instance `Q1[person]`.

The bridge is one-way. An entry here may depend on entries of `definitions.md` — its texts, sets, functions, and numbers are how organisational facts get mechanical carriers — but no entry of `definitions.md` may ever depend on an entry here. `definitions.md` stays self-contained; this ledger is built on top of it.

What this ledger must respect. Per `../.. / RULES.md`'s clarification that defines *organisational*: carriers translate, bindings do not. Where an entry reaches a binding — the point at which an artifact is tied to a person by nothing further — that binding is a primitive here, stated honestly, not dressed up as a mechanism.

Candidates

From the translation's residue, worked through under the protocol. The list is empty: every candidate has become an entry or been resolved below.

Resolved without becoming entries: *identity* — constituted by `Q3[succession]` rather than defined, since two texts are states of one artifact exactly by being items of one succession; *pastness* — being an earlier item of a history (`O4[version]` Notes); *hierarchy* — the arrangement of delegation viewed as a comparison on persons and groups, whose properties (transitivity, tree-shape) belong to the organisation as `P5[less than]`'s order properties belong to the set; *reporting* — `Q4[obligation]` with a delivery-shaped description, a composite in a role; *accountability* — `Q1[person]`'s second clause is the capacity and `Q2[attribution]` its assignment, so the whitepaper's "accountable to a named human" is `O2[owner]` with a non-empty group; *inheritance* — mechanically, text composition with union of declared checks; organisationally, scope nesting (`O6[scope]` Notes); *trust* — expectation that the bindings hold — attribution, succession, compliance — grounded in carriers and history: the whitepaper's outcome word, never its mechanism, sayable in the ledger's vocabulary whenever needed.

Primitives

Q1 — person

Assumed meaning. Someone who can act and be answerable for what they did. **Why left undefined.** Any definition — a legal person, a rational agent, a bearer of rights — imports a body of theory larger than this ledger to remove one word. “Someone who can act and be answerable” is reliably the same idea for every reader. **Added.** 2026-08-12 **Notes.** The two clauses are both load-bearing, and the second does the discriminating work: an AI worker acts, but is not answerable — which is the entire reason the whitepaper places a named human behind every artifact, and why *accountable* cannot bottom out in a Cog. Who counts as a person in a deployment is decided by the organisation’s own context — a legal question, not this ledger’s — the same move P2[symbol] makes in letting the set decide what a symbol is.

Q2 — attribution

Assumed meaning. That a person stands behind a text: they answer for it. **Why left undefined.** Every candidate definition describes a carrier — a signature, a name written into the file, a row in a registry — and each carrier is again a text, whose tie to the person is the very thing being defined. The regress ends only at the binding itself, and the preamble requires bindings to be stated as primitives, not dressed up as mechanisms. **Added.** 2026-08-12 **Notes.** Attribution is not authorship: one can stand behind a text one did not write — approval and adoption are attributions, and the whitepaper’s Frame owners need not be its authors. It is not a function: many persons may stand behind one text and one person behind many (contrast P4[function]). A set of two-item sequences (identifier, text) can *record* attributions — that is what a registry is — but the record is a carrier, and the tie of identifier to person is this primitive again. An attribution is a fact at a moment; transfer and revocation need history, which this ledger does not yet have. A group standing behind a text *as a unit*, rather than through its members, is deliberately not covered — it becomes its own primitive if the whitepaper’s accountability turns out to need committees that answer collectively. Attribution is to the text as content, and a text is one thing wherever it occurs (P6[same as]): whoever stands behind a text stands behind its content, not behind every act of inscribing it — another’s identical inscription is another’s deed. This grain is the whitepaper’s own: an expert stands behind a Guard’s content wherever it is installed, which is what makes standing-behind exchangeable (§6.1). Its shadow is replay — a carrier over short common content binds too little context — and practice answers by writing uniqueness into the content itself, nonces and context fields: the assembly’s delimiter lesson in another key.

Q3 — succession

Assumed meaning. That the items of a sequence of texts are the states of one artifact, each holding until the next. **Why left undefined.** Two bindings in one, and neither is derivable from the texts: that the texts belong to *one* thing, and that their order is the order in which they held. Any evidence offered for either — a log, a set of timestamps, a chain of hashes — is another text, whose faithfulness is the thing being assumed. The regress ends at the binding, as with Q2[attribution]. **Added.** 2026-08-12 **Notes.** Stated in proposition form so that definitions can use it as a condition — the move P5[less than] and

P6[same as] make in the mechanical ledger. Not the ordinary English “in succession”, which means merely *consecutively* and is true of every sequence: what is assumed here is that the items are states of **one artifact**, in the order those states held. Identity through change needs no separate entry, but not because it is derived: “one artifact” is part of what this primitive assumes, and the ledger has no access to “same artifact” except through membership in one succession. A repository does not contain an artifact plus its history; the history is the artifact.

Q4 — obligation

Assumed meaning. That a person must bring about what a text describes, and answers for it if it is not brought about. **Why left undefined.** The deontic *must* cannot be built from what either ledger holds: it is not a text (writing “must” obliges no one), not a function, and not an attribution — a person can stand behind a text without being obliged by it. Every candidate definition — a rule written somewhere, an agreement, a threat of consequence — is a carrier whose bindingness is the thing being defined. The regress ends at the binding, as with Q2[attribution] and Q3[succession]. **Added.** 2026-08-12 **Notes.** Operands: a person and a text, the text *describing* what must be brought about — and the reading of the description is a convention outside both ledgers, exactly as “determines” is for D10[model specification]. Whether what was done satisfies the description is a further binding — call it compliance — settled by neither the deed nor the text. The second clause ties the notion to Q1[person]: an obligation is what makes non-performance something to answer for. A group is obliged through its members, as with Q2[attribution]; collective obligation is deferred on the same trigger. A prohibition is an obligation whose described state is an absence — that no private data is released — carried by the description as all content is; the whitepaper’s Policy Guards are mostly of this shape.

Q5 — permission

Assumed meaning. That a person may bring about what a text describes: doing so is not, by itself, something to answer for. **Why left undefined.** The tempting definition — permitted is what is not forbidden — needs negation over deeds and a closed catalogue of the forbidden, machinery neither ledger has; and in organisations permission is not the absence of prohibition anyway but the presence of a grant, which presupposes this notion rather than defining it. **Added.** 2026-08-12 **Notes.** The duality with Q4[obligation] is deliberately routed through answerability, not negation: obligation makes *not doing* answerable; permission makes *doing* unanswerable. (This is the shape of Hohfeld’s duty and privilege — an outside anchor, named as outside.) Permission covers the deed, not its manner: how something permitted was done can still be answered for. Granting and revoking — bringing it about that another person is permitted — are deeds like any other, described by texts about permissions; that self-reference is what makes delegation definable rather than primitive.

Definitions

O1 — group

Definition. A set of persons. **Depends on.** P3[set], Q1[person] **Added.** 2026-08-12

Notes. The first bridge crossing: the set is the mechanical ledger’s, the members are this

ledger's. A group of one is a group, so the whitepaper's recurring "a human or a group of humans" (§4.3) collapses to "a group" with no loss. Whether a group of none exists is the zero-items question again, deliberately open in both ledgers; nothing here creates one, and an empty owner — a text no one is behind — is exactly the situation the whitepaper exists to prevent for its governed artifacts.

O2 — owner

Given. a text *t* **Definition.** A group, each of whose members stands behind *t*. **Depends on.** D2[text], O1[group], Q2[attribution] **Added.** 2026-08-12 **Notes.** Because every ship-pable artifact is a text, one definition covers the whitepaper's four marketplace classes at once — Frames, Guards, Op manifests, model specifications — and the Hub's registry rows besides. Nothing here gives every text an owner, and that is by design: most texts — a typed prompt, a generated output — have no one behind them, and should not. Should the group of none be admitted, it qualifies vacuously, and the empty owner reads as ownerlessness: "t's only owner is empty" and "no one stands behind t" say the same thing, so the zero-items question has no stake here. The whitepaper's requirement that its artifacts be owned — §4.3's *Owned* row, §6.1's "named, accountable owner" — is a policy about those texts, enforced in carriers such as registries, and is therefore checkable rather than true; the paper's own gloss supports answerability as the core, pricing an artifact by "who stands behind it". Of the deferrals this entry once carried, two have discharged: *management* — the right to change the text and gatekeep changes (§7.5) — is Q5[permission] over that deed, and *transfer* is statable through O3[history] and O4[version], the owner of one version and of the next being free to differ. What remains is *uniqueness*: nothing makes an owner *the* owner; that a text has exactly one is policy.

O3 — history

Definition. A sequence of texts whose items stand in succession. **Depends on.** D2[text], P1[sequence], Q3[succession] **Added.** 2026-08-12 **Notes.** Most sequences of texts are not histories, and whether one is cannot be read off the sequence — the restriction is Q3[succession]'s binding, assumed rather than checked; a carrier such as a hash chain can make forging a *record* of succession costly, but cannot make the record true. The carrier-binding split at its cleanest: the sequence is fully mechanical — a git chain, a registry's rows — and translates completely; that it faithfully records one artifact's states does not. A history is append-shaped by its meaning rather than by any rule here: rewriting an earlier item does not change what held, it produces a different record of succession — a claim, not a mechanism, and exactly what the whitepaper's audit story needs to be able to say.

O4 — version

Given. a history *h* **Definition.** An item of *h*. **Depends on.** P1[sequence], O3[history] **Added.** 2026-08-12 **Notes.** A text in a role, as D4[token] is a symbol in a role. *Earlier* and *later* among versions come free from P1[sequence]'s order; *pastness* is being an earlier item. The whitepaper's "versioned" property (§4.3, §4.5) translates as: there exists a history whose items are the artifact's states — an organisational fact, carried by whatever maintains the record. Transfer of ownership is statable with no further machinery: the owner of one version and of the next may differ.

O5 — delegation

Given. persons p and q , and a text t **Definition.** That p is permitted to bring about that q is permitted what t describes. **Depends on.** D2[text], Q1[person], Q5[permission] **Added.** 2026-08-12 **Notes.** A permission about a permission — no new primitive, as Q5[permission]’s Notes promised. Revocation is the same shape with an ending in the described deed, and the description carries that content the way all descriptions do: by convention, outside the ledgers. Nothing here makes delegation transitive or exclusive; what an organisation’s grants add up to is a property of that organisation.

O6 — scope

Definition. A text that determines a group. **Depends on.** D2[text], O1[group] **Added.** 2026-08-12 **Notes.** The criterion, not the collection: the determined group varies as people come and go while the scope stays the same text — exactly as D10[model specification]’s set of models varies by machine while the specification stands. Determination rests on a reading convention plus facts on the ground, as it does for every specification. The whitepaper’s scope kinds — organization, department, team, project, role, relationship (§4.3) — are all such texts. A Frame *applies within* its scope when each member of the determined group is obliged as the Frame describes for them — Q4[obligation] composed, needing no entry of its own until something wants the name. Nesting is membership between determined groups: s lies within s' when every member of the group the one determines is a member of the group the other does. The whitepaper’s other use of the word — credentials “scoped” (§4.2) — is not this scope but a permission with a narrower describing text, and needs no entry.

O7 — provenance

Given. a history h **Definition.** That someone stands behind each version of h . **Depends on.** Q1[person], Q2[attribution], O3[history], O4[version] **Added.** 2026-08-12 **Notes.** The first defined *condition* — a proposition built from prior entries, Q2[attribution] taken across O4[version]s — which the Clarification on primitives licenses for use exactly as the primitive conditions are used. The whitepaper’s thesis now has a formal reading: “provenance is the product” (§6.1) prices this condition together with Q3[succession] itself — and both are bindings, so what a marketplace actually moves is *records claiming provenance*: the record translates completely, being texts, while the claimed fact never does. A signature chain raises the cost of forging the record; nothing establishes the fact. That is why provenance cannot be generated, cannot be faked from inside the loop, and can be the product — the whitepaper’s §6.1 argument, derived rather than quoted. Provenance here is custody, not derivation: that a version was produced from other texts is recordable mechanically — D17[run specification] holds exactly that — and is the other half of what practice calls provenance; the two halves meet in a record, and only this half is bindings.

O8 — approval

Given. a person p and a text t **Definition.** A text that p stands behind, describing a permission of what t describes. **Depends on.** D2[text], Q1[person], Q2[attribution], Q5[permission] **Added.** 2026-08-12 **Notes.** The runtime picture — halt until a human presses ok — decomposes: the halt is a computation blocking on an unsupplied input, ex-

actly as it blocks on draws, and is no part of the function; the pressed symbol is data entering where draws enter (D20[checked run] Notes); and what makes the event an *approval* is neither of these but the attribution — the bit binds no one, the text behind which p stands does, and that binding is what the whitepaper’s approval Gates exist to produce. A refusal is not the absence of an approval but its own artifact — a text p stands behind describing that the deed is *not* permitted, an absence-shaped description as in Q4[obligation]’s Notes — which is why audit practice records refusals, not just consents. Q2[attribution]’s grain applies with force: an approval must name what it approves distinctively, or it approves too much — the reason real approval records reference the artifact by hash, uniqueness written into content.

O9 — retention

Given. a person p and a history h **Definition.** That p must bring about that h is kept, and extended only in succession. **Depends on.** Q1[person], Q3[succession], Q4[obligation], O3[history] **Added.** 2026-08-12 **Notes.** The whitepaper’s `retain_for: 7_years` (§5.6) and §3.4’s retention policies. Durations are beyond both ledgers — “seven years” is carried by the description’s convention, since succession gives order and not distance. The “extended only in succession” clause is O3[history]’s append-shape made someone’s duty.

Revision log

Every change to an existing entry under R5, with the re-verification result.

2026-08-12 — migrated to the hypelysis repository — accepted

Change. The governance pointer moved from `definitions.md`’s rules sections to the repository-level `../../RULES.md`, where they now live for every study. Nothing else changed.

R5/R7. Structural, not semantic. **Passes.**

2026-08-12 — audit closures: Q3 reworded, Q4 and O7 extended — accepted

Change. Three closures from the are-you-sure audit. Q3[succession]’s Notes claimed identity through change was “constituted rather than presupposed” while the assumed meaning itself uses “one artifact” — an overclaim; the Notes now say the true, weaker thing: identity needs no separate entry because the ledger has no access to “same artifact” except through membership in one succession. Q4[obligation] gains the prohibition reading — an obligation whose described state is an absence — the whitepaper’s most common Guard shape. O7[provenance] declares its deliberate exclusion: custody, not derivation, the latter being mechanically recordable in D17[run specification].

The audit, closed. Thirty-nine entries across both ledgers, examined in order under one question. Outcome: P1 and P4 amended (the single item; no-value inputs outside the domain), D8/D9 corrected under R9 (the number set named), Q2’s grain declared, and the three closures above; everything else confirmed under pressure. The pattern of what surfaced: wordings that left load-bearing cases to charity, parameters carried silently, and commentary claiming more than its entry earns — never a definition wrong outright.

R5/R7. Notes are commentary throughout; no assumed meaning or definition changed beyond P1/P4/D8/D9, each logged in the mechanical ledger. **Passes.**

2026-08-12 — Q2[attribution] Notes: the grain declared — accepted

Change. Attribution’s operand is a text, and texts are abstract — one thing wherever they occur — so standing behind a text is standing behind content, not behind acts of inscription. The grain was correct and undeclared; the Notes now declare it, with the confirmation (the whitepaper’s §6.1 pricing is content-grained, which is what makes vouching exchangeable) and the shadow (replay, answered in practice by writing uniqueness into content). Surfaced by the are-you-sure audit.

R5/R7. Notes are commentary; the assumed meaning is unchanged. O2[owner] and O7[provenance] inherit the declared grain and re-read cleanly under it. **Passes.**

2026-08-12 — O2[owner] Notes: two deferrals discharged — accepted

Change. O2’s Notes deferred *management* to permission and *transfer* to history; both now exist — Q5[permission], and O3[history] with O4[version] — so the Notes record the discharge and keep only *uniqueness*, which is policy rather than missing machinery. Notes may cite later entries, as in the mechanical ledger; the core fields do not.

R5/R7. Notes are commentary; the definition is unchanged and nothing re-verifies. **Passes.**